

chapter 5

Advanced Training Practices

Traditional resistance training practices form the cornerstone of human muscle development. However, as one gains training experience, he or she can employ more advanced training practices to maximize individual genetic hypertrophic potential. These strategies generally allow increases in volume and intensity of load, primarily through the use of heavier loads, prolonging set duration, or both. In some instances, the strategies may also enhance hypertrophic mechanisms (mechanical tension, metabolic stress, and muscle damage) over and above what is possible through traditional resistance training practices.

Advanced training practices can be broadly classified into two categories. The first is *accumulation strategies* that facilitate the ability to achieve greater training volumes; examples include drop sets, supersets and pre-exhaustion, and loaded stretch. The second is *intensification strategies* that increase loading capacity; examples include intraset rest and accentuated eccentrics. The following is an overview of these strategies, which all have at least some higher-level evidence to either support or refute their use in hypertrophy-oriented resistance training programs.

Loaded Stretch Training

Stretch training is commonly prescribed for improving measures of mobility and flexibility. However, evidence indicates that forms of stretch training may in fact mediate anabolic adaptations. For instance, research shows a

hypertrophic benefit to dynamic training at long muscle lengths (i.e., stretched position) compared to training in a shortened position (52, 53). The mechanistic reasons for these findings remain unknown, but possibilities include heightened ultrastructural disruption, greater mechanical stress, or perhaps a combination of the two phenomena. Regardless of the mechanisms, a logical rationale exists for integrating stretch training into a resistance training program to confer an additive effect on muscle growth.

In vitro evidence shows that passive stretch elicits a robust anabolic response (56). However, these findings have limited in vivo applicability, and in fact such protocols generally have not demonstrated an ability to mediate long-term muscle growth in humans (1, 16). That said, a recent study suggests that the inclusion of 30-second bouts of passive stretching during each 90-second interset rest period of a traditional resistance training program may promote favorable effects on hypertrophic outcomes (20). On the other hand, passive interset stretching may negatively affect performance on subsequent sets, raising questions as to the veracity of these findings. Given the very limited evidence to date, further research is required to glean greater practical insights on the topic. Some evidence suggests that higher-intensity passive stretching may mediate alterations in fascicle length (26), although these findings seem to be relegated to the early phase of training.

The intensity of stretch training can be heightened by the addition of a load. The use

of loaded stretch for promoting hypertrophy has sound research-based support in animal models. Seminal studies from the lab of William Gonyea demonstrated that the application of a load to the stretched wings of Japanese quails produced rapid and marked increases in muscle mass. In one study, the birds' wings were chronically elevated and loaded with a weight corresponding to 10% of body mass, thereby placing the anterior latissimus dorsi muscle under persistent stretch (2). After 30 days of consistent loaded stretch, muscle cross-sectional area increased 57% and fiber number increased 52%, indicating that both hypertrophy- and hyperplasia-mediated changes had taken place. A follow-up study (5) submitted the right wings of birds to progressive stretch with loads equating to 10% to 35% of body mass for 37 days while the left wings served as unloaded controls. The initial 2 weeks involved intermittent stretch training, with each increase in load preceded by 2 to 3 days of unloading; thereafter, loading was applied daily. As previously demonstrated, large post-study changes (~300%) in muscle mass were seen, with gains attributed to both hypertrophy and hyperplasia. Although these studies provide compelling evidence that loaded stretch elicits a potent hypertrophic stimulus, the extreme nature of the protocols have little generalizability to traditional forms of resistance exercise.

To date, few human studies have endeavored to evaluate the effects of loaded stretch training on muscular development. Employing a within-subject design, Simpson and colleagues (81) had subjects perform 3 minutes of dorsiflexion of the nondominant leg on a leg press machine with a load equating to 20% of maximal voluntary contraction; loads were progressively increased by 5% weekly, with training carried out 5 days per week for 6 weeks. The authors reported significantly greater increases in gastrocnemius thickness in the stretched versus non-stretched limb. However, subsequent data provided in response to a letter to the editor of the journal (42) indicated similar post-exercise increases between stretched and non-stretched conditions (5.9% vs. 7.6%), thus calling into question the veracity of the findings.

An intriguing potential strategy for enhancing hypertrophic gains is to integrate loaded stretch into the interset rest periods. In support of this approach, Silva and colleagues (80) randomized 24 resistance-trained men to perform 4 sets of plantar flexion on a leg press machine at a load corresponding to 8RM to 12RM either with or without interset stretching. The stretch training protocol required participants to maintain the load of the machine for 30 seconds in dorsiflexion after completion of each set, whereas the non-stretched group rested passively throughout the rest period. Training was carried out twice per week for 5 weeks. Results showed post-exercise increases in muscle thickness favored the group that performed loaded stretch training compared to passive rest (23% vs. 9%, respectively). The underlying mechanisms of these findings are unclear but may be related to a greater time under tension, essentially corresponding to a higher volume or perhaps higher mechanical tension achieved by extended loading at long muscle lengths. However, it should be noted that these data have not been published in a peer-reviewed journal and thus must be interpreted with circumspection.

KEY POINT

Loaded stretch training presents an intriguing strategy for enhancing hypertrophic gains. Although evidence is still preliminary, a logical rationale exists for integrating loaded stretch training into the interset rest period. As demonstrated by Silva and colleagues (80), a viable way to implement the strategy is to complete a set and then hold the load in the stretched position for a prescribed amount of time. There is insufficient research to develop strong evidence-based guidelines for an appropriate duration of loaded stretch, and thus experimentation with different durations is warranted; anecdotally, 10 to 30 seconds seems to be a good starting point, with adjustments made based on response. Importantly, adequate rest (≥ 90 seconds) should be afforded after cessation of the stretch to prevent compromising the load lifted in the ensuing set.

Intraset Rest Training

Traditional resistance training involves the continuous performance of repetitions over the course of a set followed by a prescribed rest period to allow for adequate recovery before initiating the next set. When training in moderate to high repetition ranges, the continuous contractions result in extensive peripheral fatigue, assuming a high level of effort is employed during training. The corresponding metabolite accumulation leads to a decline in force-generating capacity, ultimately impairing the ability to sustain performance.

Intraset rest training has been proposed as a method for overcoming the negative effects of peripheral fatigue. As the name implies, the strategy involves resting for a prescribed period of time between repetitions within a given set. These in-traset rest periods conceivably allow for the accumulation of a greater total training volume while maintaining high magnitudes of loading, which in turn may promote superior muscular adaptations. However, given the mechanistic role of metabolic stress in resistance training-induced hypertrophy (86), it is unclear whether potential benefits associated with in-traset rest supersede the negative implications of altering fatigue levels.

In-traset rest training (also called cluster sets or rest-pause training) is essentially a catch-all phrase; there are myriad ways to carry out the strategy from a practical standpoint. General recommendations for this strategy prescribe rest intervals of 10 to 30 seconds between repetitions (39), although no guidelines exist for the point at which these rest intervals should be implemented during the course of a set. Moreover, the strategy can be performed in an undulating manner in which resistance is progressively increased in a pyramid-type fashion or as an ascending cluster set in which resistance is increased on each successive repetition (39).

Several studies have investigated the acute responses of resistance training performed with in-traset rest. A consistent finding is that the strategy increases markers of volume (e.g., repetition volume, volume load) compared

to training in a traditional manner (41, 44, 62). Given the well-established relationship between volume and hypertrophy (76), this indicates a potential benefit to inserting in-traset rest periods during training. Moreover, the inclusion of in-traset rest periods may allow for the use of greater external loads compared to traditional training (84) and thus amplify mechanical tension, which in turn could elicit a heightened hypertrophic stimulus (86).

In an effort to determine whether these theoretical benefits translate into a greater anabolic response, a recent study compared the acute myokine release between traditional resistance training and an in-traset rest protocol consisting of 4 sets of the back squat at 70% of 1RM in resistance-trained men (63). A crossover design was employed in which participants performed both protocols separated by a 1-week washout period. For the traditional resistance training protocol, participants performed 10 repetitions with 2 minutes rest between sets; the in-traset rest condition involved performing the first 5 repetitions of each set, taking a 30 second rest, and then completing the final 5 repetitions. Results showed that only the traditional scheme produced elevated levels of interleukin-15 (IL-15) levels at 24 and 48 hours post-exercise. Given that IL-15 has been implicated as a mediator of muscle mass (59), these findings suggest that the use of in-traset rest may be suboptimal for maximizing hypertrophy, at least when performed with 30-second in-traset rest periods.

There is a relative paucity of longitudinal research on the hypertrophic effects of in-traset rest training. Oliver and colleagues (61) randomized 22 resistance-trained men to a total-body resistance training program performed using either a traditional or a cluster set protocol. The traditional group performed 4 sets of 10 repetitions with 2 minutes of rest between sets, whereas the cluster set group performed 8 sets of 5 repetitions with 60 seconds of rest. Training was carried out 4 days per week for 12 weeks, with volume load equated between conditions. Post-exercise body composition measures showed greater absolute gains in lean mass for traditional compared to cluster set training (2.3 vs. 1 kg, or 5.1 vs. 2.2 lb, respec-

tively), although results did not reach statistical significance.

In the only current study to employ site-specific measures of muscle growth, Prestes and colleagues (69) randomized 18 resistance-trained individuals to perform 18 repetitions in multiple exercises for major muscle groups of the body using either a traditional resistance training protocol or a rest-pause protocol. The traditional training group performed 3 sets of 6 repetitions at 80% of 1RM with 2- to 3-minute rest intervals between sets, whereas the rest-pause group performed the first set to muscle failure at an intensity of 80% of 1RM, rested 20 seconds, and then performed additional repetitions interspersed with 20-second rest intervals until completion of the 18 repetitions; 2- to 3-minute rest periods were afforded between exercises. Training was carried out 4 days a week for 6 weeks. Pre- to post-study changes in muscle thickness of the quadriceps, as measured by B-mode ultrasound, were significantly greater in the rest-pause group compared to the traditional group (11% vs. 1%, respectively). No significant between-group differences in thickness of the arm and chest muscles were noted, but effect size increases again favored the rest-pause condition. The study design was limited by the fact that the rest-pause group trained with a higher level of effort compared to the traditional group, which may have confounded results.

Drop Sets

Drop sets (also called descending sets or breakdown sets) are one of the most popular advanced training strategies for enhancing muscle growth. The approach involves carrying out a set to concentric muscle failure and then, with minimal rest, performing as many repetitions as possible at a reduced load. The magnitude of reduction generally ranges from 20% to 25% of initial load (4, 23, 25), although higher or lower percentages can be employed because no accepted guidelines exist. If desired, double or triple drops can be implemented to elicit greater motor unit fatigue.

Claims for a hypertrophic benefit of drop set training are based on the theory that training “beyond” muscle failure can elicit a heightened stimulation of the working musculature. Specifically, muscles are not fully fatigued when sets are taken to concentric muscular failure at a given load because they are still able to produce force at lower loads. Thus, it is conceivable that performing additional repetitions at a reduced load immediately after achieving muscle failure in a set may facilitate greater fatigue of muscle fibers and, in turn, enhance the anabolic response (73). Moreover, it is conceivable that prolonging time under load may also promote an additive hypertrophic stimulus. In particular, the sustained compression of vessels can heighten local ischemia, which has

KEY POINT

It is not clear whether the use of intraset rest training is a viable strategy for enhancing the hypertrophic response to resistance training. Although it appears the increase in training volume may translate into greater gains in muscle size, the reduction in constant tension and corresponding metabolite accumulation may counteract any potential benefits and possibly have a negative effect on hypertrophic outcomes (36). As noted, longitudinal research is limited, and the existing data is conflicting, precluding the ability to draw strong conclusions into potential benefits or detriments of intraset rest. The differences in the duration of intraset rest periods further clouds the ability to infer causality.

It can be speculated that the best use of intraset rest training may be to implement it in a manner similar to the protocol of Prestes and colleagues (69), whereby a set is carried out at a high level of effort and short rest periods (~10 to 20 seconds) taken between repetitions. From a practical standpoint, it is important to completely unload during the rest phase so that sufficient recovery is attained. For example, in exercises such as the squat or bench press, the lifter should rerack the weights to prevent unwanted fatigue from isometrically holding the load, which would impair performance on subsequent repetitions.

been implicated in resistance training–induced increases in muscle mass (36).

Early support for drop set training was derived from evidence that performing a low-intensity set (50% of 1RM) of knee extensions immediately after a set carried out at 90% of 1RM resulted in significantly higher post-exercise elevations in growth hormone compared to performing the high-intensity protocol alone (32). However, emerging research calls into question the extent to which acute hormonal fluctuations mediate hypertrophic adaptations, raising doubt about the relevance of these findings. Recently, it was demonstrated that drop set training heightened motor unit activation and intramuscular hypoxia in trained lifters, whereas these effects were not seen in untrained individuals (35).

The results of several longitudinal studies provide interesting insight into the effects of drop sets on muscle growth. Seminal work by Goto and colleagues (33) indicated potential benefits to incorporating drop sets into traditional heavy-load resistance training programs. A cohort of recreationally trained men initially performed a 6-week hypertrophy-oriented lower-body routine (leg press and leg extension) that resulted in a 4% increase in thigh muscle cross-sectional area. The participants were then randomly assigned to a group that performed either 5 sets of lower-body exercise at 90% of 1RM or the same routine with the addition of a drop set at 50% of 1RM. Following 4 weeks of the extended protocol, those in the drop set group continued to see increases in thigh muscle cross-sectional area (~2%), whereas the group performing only high-intensity training showed a slight decrease in muscle size (~0.5%). Although the positive hypertrophic effects observed for drop set training are intriguing, it should be noted that the drop set group performed a higher total training volume, which may have confounded results.

Subsequently, Fisher and colleagues (25) randomized a cohort of resistance-trained men and women to a single-set resistance training program using one of the following three conditions: (1) a load of 8RM to 12RM, (2) a load of 8RM to 12RM with an added drop set at a 30% reduction of the initial training load, or

(3) a load of 4RM followed by two drop sets with successive reductions of 20% of load. The program employed exercises for all the major muscle groups; however, drop sets were performed for only the lat pulldown, chest press, and leg press. After 12 weeks, post-exercise changes in fat-free mass showed no significant differences between conditions despite a greater volume performed by the drop set groups. A limitation was the measurement of fat-free mass by air displacement plethysmography (e.g., BodPod), which provides only a gross estimate of all nonfat components (e.g., muscle, bone, body water); thus, results cannot necessarily be extrapolated to changes in skeletal muscle size. This is particularly pertinent given that drop sets were performed on just three of the exercises.

Employing a within-subject design, Angleri and colleagues (4) randomized the legs of resistance-trained men to either a traditional set or drop set protocol of leg press and leg extension exercise with volume load (sets $\times$ repetitions $\times$ load) equated between conditions. The leg assigned to traditional training performed 3 to 5 sets of 6 to 12 repetitions with a 2-minute rest interval between sets; the other leg performed the same routine with up to 2 drop sets added at a 20% reduction in load. At the conclusion of the 12-week study period,

KEY POINT

The inclusion of drop sets in a resistance training program is of questionable benefit to hypertrophy when total session volume is equated. However, drop sets may be useful for increasing total training volume without substantially increasing session duration. This not only makes workouts more efficient, but it can also potentially reduce fatigue occurring from extended-duration training sessions. Given the need to train to failure when employing drop sets, it may be best to limit the strategy to the last set of a given exercise. Weight stack machines are particularly well-suited to drop set training because loads can be quickly decreased simply by moving a pin.

PRACTICAL APPLICATIONS

COLD-WATER IMMERSION: HYPERTROPHIC FRIEND OR FOE?

Proper recovery from training is considered essential to optimizing muscle gains. Various passive techniques have been advocated to enhance the post-exercise recovery process and restore the body to its normal physiological and psychological state. Cold-water immersion is one of the most commonly used modalities in this regard. The technique involves immersing all or part of the body in cold water (figure 5.1). The specific protocols vary, but prescriptions generally include water temperatures cooler than 15 °C (59 °F), with immersion durations of at least 10 minutes (12).

The findings of several systematic reviews and meta-analyses indicate that cold-water immersion helps to attenuate delayed-onset muscle soreness (DOMS) (8, 9, 46). Given that DOMS may impede lifting performance, the use of cold-water immersion seems to be of potential benefit for those involved in intensive resistance training programs. However, despite its potential recovery-related benefits, emerging evidence indicates that cold-water immersion may be detrimental to muscle development.

Acute research shows that cold-water immersion impairs intracellular anabolic signaling, and an attenuated p70^{S6K} phosphorylation response is seen over the course of a 48-hour recovery period after resistance training compared to an active recovery period (72). The same study also showed cold-water immersion mitigated the number of Pax7+ cells and NCAM+ cells at 24 and 48 hours after resistance exercise, indicating a deleterious effect on the satellite cell response to resistance training as well. Other research shows cold-water immersion suppresses ribosome biogenesis (21), which is thought to be a key player in the long-term regulation of muscle growth (22).

The negative acute effects on anabolism seen with cold-water immersion align with findings of longitudinal research on hypertrophic outcomes. Roberts and colleagues (72) investigated the impact of cold-water immersion versus active recovery during a 12-week resistance training program. Twenty-four physically active young men were randomly assigned to one of the two recovery conditions. Cold-water immersion was initiated within 5 minutes post-exercise and involved sitting waist deep in water approximately 10 °C (50 °F) in an inflatable bath for 10 minutes. Those in the active recovery group cycled on an ergometer at a self-selected low intensity for 10 minutes. Results demonstrated a blunting of both whole-muscle hypertrophy and histological measures of Type II fiber cross-sectional area with the use of cold-water immersion. Similar findings were observed by Yamane and colleagues (90), who compared muscular adaptations between cold-water immersion and passive rest following a 6-week resistance training program of the wrist flexor muscles. Cold-water immersion treatment consisted of immersing the trained arms in a water



FIGURE 5.1 An athlete in cold-water immersion.
AI Powers/Zuffa LLC/Zuffa LLC via Getty Images

(continued)

Cold-Water Immersion: Hypertrophic Friend or Foe? *(continued)*

bath within 3 minutes post-exercise. Water temperature was maintained at 10 °C (50 °F), with immersion lasting for 20 minutes. Results showed that both groups increased thickness of the wrist flexors, but hypertrophy was significantly greater in the passive recovery condition.

The underlying mechanisms by which cold-water immersion impedes anabolism remain unclear. Given that the acute inflammatory response to resistance training is implicated in anabolic signaling (75), and given that cold-water immersion alleviates symptoms of DOMS, which is associated with induction of acute inflammation, it would be logical to speculate that the anti-inflammatory effects induced by cold-water immersion play a mechanistic role. However, research on the topic is somewhat contradictory. Peake and colleagues (67) found that cold-water immersion did not alter post-exercise levels of proinflammatory cytokines and neurotrophins, or intramuscular translocation of heat shock proteins compared to active recovery following resistance training. Alternatively, Pournot and colleagues (68) reported a diminished inflammatory response when cold-water immersion was applied pursuant to an endurance exercise bout.

It can be hypothesized that negative anabolic effects of cold-water immersion are in some way related to a reduction in blood flow (50, 54), possibly by compromising post-exercise amino acid delivery to muscle (29). Exposure to cold temperatures also has been shown to interfere with anabolic signaling, potentially via an upregulation of AMPK, a known inhibitor of mTOR (12). Because research is limited, it is difficult to draw strong inferences on the topic.

In summary, current evidence contraindicates the use of cold-water immersion for those seeking to maximize muscular development, at least when used regularly. Any benefits to recovery seem to be outweighed by an impaired anabolic response to resistance training. Post-exercise heat therapy represents a promising strategy for enhancing recovery without interfering with hypertrophic outcomes and possibly even improving subsequent strength-related performance (13). Evidence shows that heat therapy, applied 2 hours daily over 10 days of immobilization, can attenuate skeletal muscle atrophy (38). Although this finding indicates potential anabolic effects, results cannot be extrapolated to benefits during performance of muscle-building protocols. Another study demonstrated that topical application of heat via a heat- and steam-generating sheet for 8 hours per day for 4 days a week significantly increased quadriceps hypertrophy over a 10-week treatment period (34). Research into this modality is preliminary and warrants further study.

similar increases in quadriceps cross-sectional area were noted between conditions (7.8% for the drop set group and 7.6% for the traditional training group), indicating no hypertrophic benefit for drop set training. Similar findings were observed by Ozaki and colleagues (64), who randomly assigned nine untrained men to perform biceps curls in one of three conditions: (1) 3 sets of heavy-load resistance training

(80% of 1RM) with a 3-minute rest period; (2) 3 sets of light-load resistance training (30% of 1RM) with a 90-second rest period, or (3) a single-set of heavy-load resistance training (80% of 1RM) followed by 4 drop sets at 65%, 50%, 40%, and 30% of 1RM. Similar increases in cross-sectional area were seen across conditions. The study was limited by a small sample size, which compromised statistical power.

Alternatively, Fink and colleagues (23) randomized 16 recreationally trained young men to perform elbow extension exercise either in a traditional fashion that consisted of 3 sets of 12RM with a 90-second rest interval or as a single 12RM set followed by 3 consecutive drop sets with load reductions of 20%. Training was carried out twice per week for 6 weeks, with total training volume equalized between conditions. No statistically significant differences in triceps brachii cross-sectional area were noted between groups; however, the relative differences in hypertrophy (10.0% vs. 5.1%) and effect size (0.22) favored the drop set condition versus traditional training, raising the possibility of a type II statistical error (i.e., false negative).

Supersets and Pre-exhaustion

Superset training refers to the performance of sets of two exercises back to back, with minimal rest between sets. Supersets can be performed in several configurations. In paired-set training, the exercises involve an agonist and antagonist (e.g., triceps pushdowns followed by biceps curls). Staggered supersets include exercises for muscles of different areas of the body (e.g., elbow flexors and plantar flexors). Compound supersets include exercises for the same muscle group (e.g., leg extensions followed by back squats). Compound supersets also can be performed to pre-exhaust the target musculature. For example, a lateral raise can be performed immediately before a military press to pre-exhaust the middle deltoid, which conceivably places greater mechanical stress on the target muscle during the ensuing multi-joint movement.

A substantial body of research has been conducted into the effects of superset training on muscle activation. Electromyography (EMG) studies carried out on paired-set training show similar activation levels compared with traditional training protocols for the upper-body musculature (71), whereas a beneficial effect has been demonstrated on activation of the lower-body musculature (48).

Compound supersets targeting the chest musculature with two multi-joint exercises (bench press followed by incline press) were found to negatively affect EMG amplitude of the clavicular head of the pectoralis major (88). EMG studies on pre-exhaustion compound supersets have produced somewhat contradictory findings. A recent study found that pre-exhausting the chest musculature with a dumbbell fly before the bench press significantly increased activation of the pectoralis major, anterior deltoid, triceps brachii, and serratus anterior versus performing the bench press alone (83). Conversely, Gentil and colleagues (30) reported that performance of the pec deck fly immediately before the bench press did not alter activation of the pectoralis major compared to performing the exercises in the opposite order, but EMG amplitude was higher in the triceps brachii when employing the pre-exhaustion technique. Yet other studies show no differences in muscle activation between pre-exhaustion compound supersets and traditional training (10, 31). Studies investigating the effects of pre-exhaustion compound supersets on the lower-body musculature have shown equally conflicting results. Augustsson and colleagues (7) reported that pre-exhaustion of the quadriceps via leg extension exercise blunted subsequent activation of the rectus femoris and vastus lateralis during the leg press. Alternatively, Rocha-Júnior and colleagues (43) found that performing leg extension exercise before the leg press resulted in a greater activation of the vastus lateralis compared to performing the leg press alone.

Several studies have investigated muscle activation during performance of supersets that involve different but related areas of the body (triceps extension and bench press): Some report greater activation of the pectorals compared to a traditional training protocol (37), while others show similar EMG amplitudes between conditions (82, 88). Overall, it is difficult to reconcile the divergent findings between EMG studies on superset training; as such, conclusions about practical implications remain equivocal.

Superset training can alter training volume, which in turn can affect hypertrophy outcomes.

Research indicates that paired-set training of the upper-body musculature (bench press and lat pulldown) produces a greater per-session volume load and a greater level of muscular fatigue than performing the same exercises in a traditional manner (65, 70). This suggests that the use of supersets performed in agonist–antagonist fashion may promote an enhanced hypertrophic training stimulus. However, other research shows no differences in volume load using the same upper-body exercises (71). Moreover, upper-body compound supersets employing two multi-joint movements (bench press and incline press) seem to have a detrimental effect on volume load (88).

Evidence suggests that the performance of supersets may have a negative impact on markers of recovery. This has been demonstrated both in compound supersets (11) as well as staggered supersets for different body regions (i.e., alternating upper- and lower-body exercises) (89) when compared to performing the same exercises in a traditional fashion. It should be noted that these findings are specific to an isolated bout of training. Given the well-established existence of the repeated bout effect in which the neuromuscular system adapts to an unaccustomed bout of exercise by becoming progressively more adept at withstanding ultra-structural damage to muscle tissue in future bouts of the same exercise (51), it is possible if not likely that the response to repeated use of such training practices would positively alter recovery capacity over time. Thus, practical conclusions on the matter should be interpreted with circumspection.

To date, only two published studies have endeavored to compare longitudinal hypertrophic adaptations in a variation of superset training versus a traditional training scheme (24). A cohort of 39 resistance-trained men and women were randomly assigned to perform a total-body resistance training program under one of three conditions: (1) perform exercises as pre-exhaustion compound supersets (e.g., pec fly before chest press, leg extension before leg press, and pullover before lat pulldown), (2) perform the same exercises in the same order, but take a 60-second rest interval between sets, or (3) perform the multi-joint exercise before

the single-joint movement for each body part with a 60-second rest interval between sets. A single set of up to 12RM was performed for each exercise, with training carried out twice per week. After 12 weeks, no significant changes in lean mass were observed in any of the conditions. Conclusions are confounded by the use of air displacement plethysmography to assess lean mass, which provides limited insights into changes in skeletal muscle hypertrophy (see chapter 3). Moreover, the low-volume training protocol may have been insufficient to produce detectable changes in lean mass regardless of measurement precision.

Most recently, Merrigan and colleagues (55) randomized recreationally active women to perform 3 to 4 sets of the squat and leg press either as compound sets or using a traditional approach. For the compound sets, subjects performed the squat and leg press in immediate succession, then rested for 140 to 150 seconds before performing the next compound set; the traditional group performed sets of squats and then the leg press with 1-minute rest intervals. Increases in muscle thickness and cross-sectional area were similar between conditions, indicating that compound sets neither hinder nor enhance muscular adaptations. It should be noted that the design of this study employed

KEY POINT

Although superset training remains a popular strategy in bodybuilding circles, purported hypertrophic benefits are based largely on anecdote. The paucity of longitudinal research on superset training precludes the ability to draw conclusions about its efficacy for enhancing hypertrophy with any degree of confidence; the contradictory evidence of its effects on muscle activation further clouds inferences. There is evidence that superset training can enhance workout efficiency and thus reduce training time (89). The strategy therefore may be a viable option when an individual is pressed for time and needs to get in a quick workout without compromising target training volume (66).

two multi-joint exercises; given that compound sets are typically carried out by combining single-joint and multi-joint exercises, results cannot necessarily be extrapolated to such a strategy.

Eccentric Overload Training

Eccentric actions, in which activated muscles are forcibly lengthened, afford the ability to use maximal loads that are 20% to 40% greater than that of concentric actions. This phenomenon has led to speculation that eccentric overload training, employing loads greater than those used during concentric actions, may provide an added anabolic stimulus.

Several lines of evidence indicate that eccentric training plays an important and potentially additive role in resistance training–induced hypertrophy. For one, lengthening actions have been shown to elicit a more rapid rise in muscle protein synthesis and promote greater increases in anabolic signaling and gene expression than other muscle actions do (17, 57, 78). There also is evidence that eccentric actions induce preferential recruitment of Type II myofibers (58), which in turn may facilitate their growth by an increase in mechanical tension. Moreover, regional increases in hypertrophy of the quadriceps are different between actions, with eccentric training eliciting greater growth in the distal portion of the muscle and concentric training promoting greater growth at the midpoint (77). Thus, the inclusion of eccentric exercise should help to promote more symmetrical muscle development, at least in the thigh musculature. Finally, it is well established that greater muscle damage occurs during eccentric actions; although speculative, this may enhance hypertrophic adaptations over time, perhaps by fostering greater satellite cell and myonuclear accretion (86).

Several studies have endeavored to compare hypertrophic changes between regimented eccentric overload training and traditional resistance training. Early work on the topic was carried out by Friedmann and colleagues (27), who randomized untrained subjects to perform low-load knee extension resistance training with the eccentric component performed either

at 30% of concentric 1RM (traditional training) or 30% of eccentric 1RM (eccentric overload); both groups performed the concentric action at 30% of concentric 1RM. After 4 weeks, the eccentric overload condition showed superior increases in muscle cross-sectional area compared to traditional training. In a follow-up study in resistance-trained men, 30 male athletes were randomly allocated to perform either traditional concentric/eccentric knee extension training or a concentric/eccentric overload routine for 6 weeks (28). As in the previous study, training for the traditional condition was carried out on a conventional device, whereas eccentric overload was performed on a computer-driven device. Results showed that whole-muscle hypertrophy was similar between conditions; however, histological analysis indicated greater Type IIx fiber growth with eccentric overload compared to traditional training. A confounding issue with both studies was the use of different training devices between conditions: Traditional training employed a conventional resistance machine, while eccentric overload training involved a computer-driven device. How this may have affected outcomes is unknown.

Horwath and colleagues (40) randomly assigned 22 resistance-trained men to perform lower-body resistance exercise consisting of either isokinetic training combined with eccentric overload or traditional isotonic training for 8 weeks. Exercises consisted of heavy squats and jump squats. Mean post-exercise increases in muscle thickness across the quadriceps muscles favored the eccentric overload training compared to traditional training (8.9% vs. 2.1%, respectively). Although these findings suggest a potential benefit to incorporating eccentric overload training into resistance training prescription, the study was confounded by the use of different modalities (isokinetic versus isotonic training). Moreover, eccentric overload was performed only on the power-oriented exercise (jump squat), which would not be an ideal strategy for inducing a maximal hypertrophic stimulus.

In what is perhaps the most ecologically valid study on the topic, Walker and colleagues (87) recruited resistance-trained men to perform

3 sets of the leg press, leg extension, and leg curl exercises on standard resistance training equipment with a load corresponding to 6RM to 10RM. Subjects were randomized to carry out training either in a traditional fashion that used a constant load for both the concentric and eccentric actions, or with eccentric overload where a 40% greater load was imposed on the eccentric portion of the lift via the use of custom weight releasers. Results showed similar overall increases in vastus lateralis cross-sectional area. There did appear to be regional hypertrophic differences between conditions, with traditional training showing greater growth at 33% of femur length and eccentric overload displaying greater growth at 50% of femur length; however, the magnitude of these differences was rather modest and within the error variance of the measurement, raising skepticism about their practical meaningfulness.

Flywheel training devices represent an intriguing tool for promoting eccentric overload (figure 5.2). These devices consist of a flywheel connected to a rotating shaft. The lifter initiates a concentric action that unwinds the flywheel's strap, which in turn transfers kinetic energy to the flywheel; this in turn requires applied force to slow movement on the eccentric action. Assum-

KEY POINT

Eccentric overload appears to be a promising strategy for enhancing muscle growth. Eccentric overload can be fairly easily employed on various machine-based exercises by performing the concentric action bilaterally and the eccentric action unilaterally. For example, during the leg curl, both legs lift the weight, and the weight is then lowered with one leg, alternating between right and left limbs on each successive repetition. On free-weight exercises, a spotter can help lift a supramaximal load after the trainee lowers it under control. A flywheel device provides an appealing option for inducing eccentric overload in a safe and efficient manner. Although no definitive evidence exists to prescribe an ideal cadence for the eccentric action, a 2-second tempo seems to be enough to ensure the load is lowered under sufficient control to bring about desired results (6, 79).

ing a high level of effort is employed, maximal forces are generated concentrically accompanied by supramaximal forces eccentrically (85). Several studies have endeavored to investigate the

hypertrophic effects of flywheel training compared to traditional resistance training. Maroto-Izquierdo and colleagues (49) randomized 29 resistance-trained men to perform 4 sets of 7 repetitions on either a standard machine leg press or a flywheel-based unit that applied eccentric overload. After 6 weeks of twice-weekly training, thickness of the vastus lateralis was significantly greater using eccentric overload at the proximal, mid, and distal portions of the muscle compared to the traditional approach. Similarly, Norrbrand and colleagues (60) found that 5 weeks of coupled concentric–eccentric flywheel training on a knee extension apparatus produced superior increases in



FIGURE 5.2 An athlete using a flywheel device for eccentric overload training.

Deadlift using kBox4, photo courtesy Exxentric.

PRACTICAL APPLICATIONS**STRATEGIES FOR REDUCING DELAYED-ONSET MUSCLE SORENESS**

As discussed in chapter 2, delayed-onset muscle soreness (DOMS) manifests 24 to 48 hours after performance of intense exercise and tends to be most prevalent when the stimulus is unaccustomed. While mild DOMS generally is benign from a performance standpoint, moderate to severe levels of soreness can impair subsequent strength capacity, thereby potentially having a detrimental impact on muscular adaptations.

Numerous strategies have been proposed to help alleviate the negative consequences of DOMS (15). A recent meta-analysis on the topic found that massage therapy had the greatest effect on reducing symptoms of DOMS, conceivably by increasing blood flow and diminishing edema. Other strategies shown to have a positive impact include compressive garments, cryotherapy, cold-water immersion, contrast water therapy (i.e., alternating hot- and cold-water baths), and active recovery. Interestingly, evidence did not show a beneficial effect of stretching on DOMS, despite its popular use as a primary treatment.

Recently, foam rolling has been advocated for counteracting DOMS. Drinkwater and colleagues (14) found that 15 minutes of foam rolling for the lower-body musculature performed immediately after a muscle-damaging eccentric bout and 24, 48, and 72 hours post-exercise significantly increased the pressure–pain threshold compared to passive recovery. These findings were associated with a greater recovery from the exercise bout, as determined by an increase in countermovement jump performance. Although speculative, a higher pressure–pain threshold conceivably could be related to a reduction in soreness, thereby raising the possibility that foam rolling may be a viable recovery option.

A potential issue when interpreting research on the topic is the possibility that findings are due to a placebo effect. It is difficult to provide adequate sham treatments as a control for manipulative therapies, and subjects therefore are not adequately blinded to the given treatment. This limits the ability to conclude whether the treatment is actually responsible for beneficial effects or if results are influenced by subjects' perception of treatment.

Importantly, while reducing DOMS potentially can benefit performance, some therapies may interfere with processes beneficial to muscle development. As noted elsewhere in this chapter, evidence shows that cold-water immersion therapy negatively affects anabolic processes (21, 72) and appears to be detrimental to long-term muscle development (72, 90). Caution should therefore be used when deciding whether to use a recovery strategy to optimize hypertrophic adaptations; the potential costs and benefits of adopting a given approach must both be taken into consideration.

quadriceps hypertrophy compared to the same set and repetition scheme (4×7) performed on a standard weight stack unit in a cohort of untrained men. Alternatively, a study employing a within-subject design demonstrated similar muscle-specific hypertrophy of the quadriceps

after 8 weeks of unilateral flywheel versus weight-stack knee extension exercise; however, the flywheel condition produced results with a markedly lower volume of repetitions, suggesting a potential beneficial effect if volume is equated (47).

Conclusion

Advanced training practices offer potential opportunities for experienced lifters to maximize their genetic hypertrophic potential. Emerging research indicates the possibility for various beneficial effects of these strategies if properly integrated into program design. However, the relative paucity of evidence makes it difficult to draw strong conclusions about how to best implement these strategies.

Eccentric overload training would seem to have the most scientific support for enhancing muscle development. The amount of evidence supporting the other strategies varies; at the very least there seems to be a logical basis for their use under certain circumstances. Several other advanced training practices such as forced repetitions and accommodating resistance using chains and bands have a hypothetical rationale that raises the possibility of a benefit when integrated into a hypertrophy-oriented resistance training program. However, research is lacking on these strategies, rendering the efficacy of their use speculative from a hypertrophy standpoint.

An advanced training practice that has received little research attention to date is inter-set isoholds. The strategy involves performing an isometric contraction of the agonist muscles immediately after the conclusion of a set. Our lab recently carried out a longitudinal study in an attempt to investigate the topic (74). Twenty-seven resistance-trained men performed a total-body resistance training program lasting 8

weeks. Training consisted of 3 sets per exercise carried out at 8RM to 12RM with a 2-minute rest period between sets. Participants were randomly assigned to perform the routine either in a traditional fashion or with an isometric isohold employed for the initial 30 seconds of each inter-set rest period. Results indicated that the isohold elicited greater increases in muscle thickness for the rectus femoris but not for other muscles of the limbs. A possible explanation may be related to the fact that the lower-body routine consisted of just the squat and leg press, which have been shown to promote preferential hypertrophy of the *vasti* muscles (vastus lateralis, vastus intermedius, and vastus medialis) at the expense of the rectus femoris (45). Alternatively, the leg extension is known to better target the rectus femoris (3, 19) and enhance its development over the course of regimented resistance training (18). It is therefore conceivable that results are a function of the specifics of the lower-body isohold protocol, which involved performing isometric holds of the quadriceps in the seated position (similar to the finish position of the leg extension exercise). This is the first and only study to date on the topic; thus, the strategy warrants further research.

Finally and importantly, some advanced training strategies can be highly taxing to the neuromuscular system. Thus, prudence is warranted if implementing them on a regular basis; periodizing their use is advisable to maximize benefits while reducing the potential for over-training.

TAKE-HOME POINTS

- Advanced training strategies can potentially augment hypertrophic adaptations; however, the overall paucity of evidence on the topic limits the ability to draw strong conclusions about best practices.
 - Eccentric overload training has the greatest research-based support for enhancing muscle development. However, evidence remains insufficient to develop evidence-based guidelines for the best way to implement the strategy in practice.
 - Although strategies such as supersets and drop sets generally have not been found to promote greater hypertrophic increases compared to traditional training, they can provide more efficient training alternatives without compromising muscle growth.
 - Given the lack of research on many of these strategies, personal experimentation is warranted to determine individual response within the context of a given training program.
 - Many advanced training strategies can be highly taxing to the neuromuscular system, and their persistent use may potentially hasten the onset of overtraining. A conservative approach is therefore warranted when implementing such strategies into a training program, and periodization (see chapter 8) should be considered for achieving an optimal cost/benefit ratio.
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